Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++
Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Feature Engineering, Exploratory Data Analysis, Supervised & Unsupervised Learning, Ensemble Methods, Model Evaluation, Statistical Analysis, Time-Series Analysis, Anomaly Detection
Deep Learning: PyTorch, TensorFlow/Keras, LSTM, CNN, Neural Networks
Generative AI & NLP: LLMs, RAG, Prompt Engineering, Google Gemini API, Document Q&A Applications
Data Analytics & Visualization: Power BI, Tableau, Matplotlib, Plotly
Backend & Deployment: FastAPI, Flask, Streamlit, REST APIs, Docker, Git/GitHub
Databases: SQL, SQLite, MySQL, MongoDB, Redis
Engineering Practices: Data Preprocessing, Model Validation, Reproducible Experimentation, PyTest, Test-Driven Development

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a Python-based analytics pipeline processing 2M+ NSE market records, combining price signals and financial-news sentiment to identify unusual market behavior.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for market anomaly detection.
- Optimized the analytics pipeline to reduce detection latency by 35% and developed interactive Streamlit and Plotly visualizations for monitoring anomalies and analyzing market trends.

***Docwise* |LINK**

- Engineered an end-to-end RAG-based document analytics pipeline using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Developed the complete document intelligence workflow from document extraction and chunking to retrieval, contextual ranking, and LLM-based response generation.
- Improved response reliability through retrieval tuning, prompt engineering, and hallucination-aware validation, increasing contextual relevance and answer consistency.

***CV-FL – Privacy-Preserving Federated Learning* |LINK**

- Designed a hybrid compression and lightweight verifiable aggregation framework for privacy-preserving federated learning in edge IoT networks.
- Investigated model-update compression, Merkle commitments, and SignScore-based verification.
- Reported approximately 19.7× compression and 100% tamper detection at 5% or greater corruption in the evaluated experiments.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025